Semester Test 2

MEMO

Electricity and Electronics EBN122

16 October 2025

Assessment Information			
Maximum marks:	46	Full marks:	45
Duration of assessment:	90 (excluding additional time)	Open/Closed book:	Closed
Additional time allocations:	5 min before and 10 min after for OCR answer sheet completion	Allowable materials:	Pencil case, non-programmable calculator
Extra-time venue:	Eng 2, 3-40	Submission format:	OCR
Total number of pages (this page included)		17	

Lecturers: Prof JP de Villiers, Mr S Twala, Prof JP Jacobs

Important

1. The examination regulations of the University of Pretoria apply.
2. The departmental rules relevant to electronically graded assessments apply.
3. Answer all questions on the OCR answer sheets provided. The question number **01** in this question paper refers to row 01 of OCR answer sheet 00/01.
4. Read the instructions on OCR answer sheet 00/01 carefully.
5. Where applicable, answers must include units.
6. Final answers must be written as real numbers and not as fractions.
7. Use **at least 3 significant digits** to write real numbers. This means that the real and imaginary parts of complex numbers should each be written using at least 3 significant digits, for example Z = 10.0 + j0.283 Ω (the same applies to magnitudes and angles).
8. Exact numbers in which decimal expansions are zero, can be written without the zero expansions, for example 0 V instead of 0.000 V or 3.5 A instead of 3.500 A.

INSTRUCTIONS

Do step 0 in the first 5 minutes of the assessment. (-5 to 0 min)

Do steps 1 and 2 in the next 100 minutes of the assessment. (0-100 min)

TIME: -5 min to 0 min

STEP 0 : Complete the front page of OCR answer sheet 00/01, and also fill in your student number on additional OCR answer sheets.

TIME: 0 min to 100 min

STEP 1: Do your calculations on the question paper.

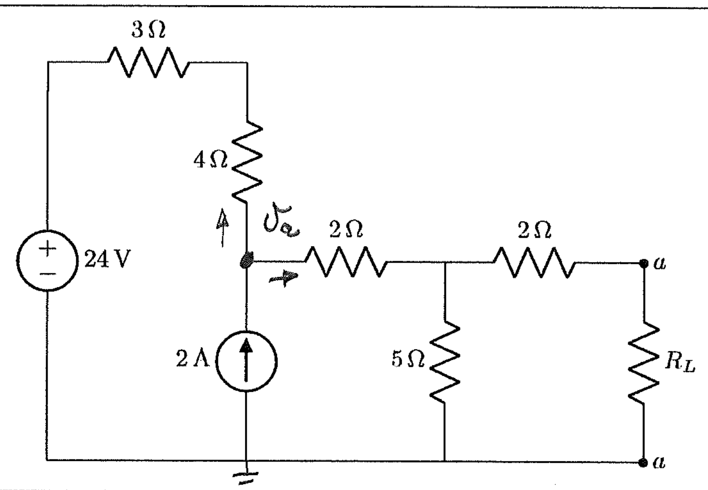
STEP 2: Carefully copy your answers to the OCR answer sheets. Be sure to include units where applicable.

- The Assessment ID is: **2025EBN122S02**
- Use a mechanical pencil with 0.5 mm HB lead to complete the form.
- Each cell should only have one character, and characters may not touch or cross the cell.
- **Do not use commas!** Please make use of full stops as decimal points.

Section 1

Questions 1 to 3 [5]

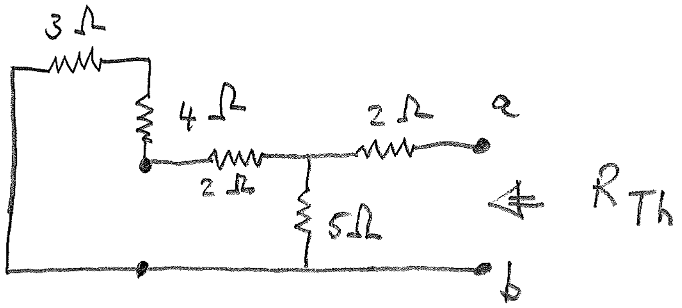
Find the Thevenin equivalent voltage V_{Th} seen by the load R_L, and the value of R_L so that maximum power is transferred to R_L. Also find p_{max}, the maximum power transferred to R_L.



01	V_{Th} [2]
02	R_L [2]
03	p_{max} [1]

01
$$\frac{v_a - 24}{7} + \frac{v_a}{7} = 2$$
$$v_a - 24 + v_a = 14 \quad \therefore \; v_a = \frac{38}{2} = 19\text{ V}$$
$$V_{Th} = \frac{5}{2+5} v_a = 13.57\text{ V}$$

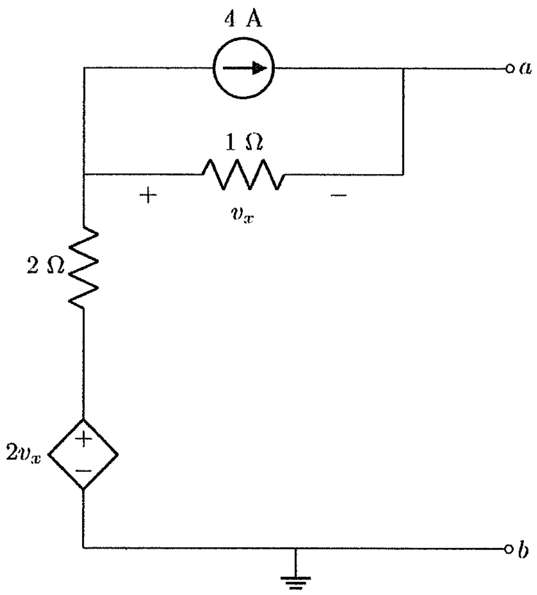
02



$$R_{Th} = 2 + 5 \,\|\, (2+4+3)$$
$$= 2 + \frac{5 \cdot 9}{5+9}$$
$$= 5.214\,\Omega$$
$$\therefore \; R_L = 5.214\,\Omega$$

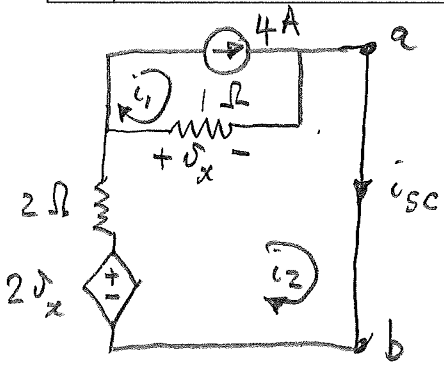
03. $$p_{max} = \frac{V_{Th}^2}{4R_{Th}} = 8.829\text{ W}$$
(continuity marking)

Questions 4 to 5 [4]



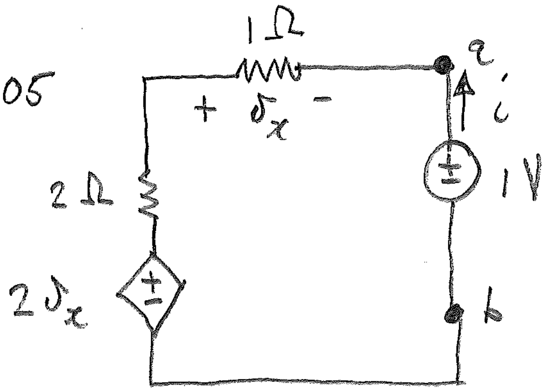
04	Determine the Norton equivalent current I_N with respect to terminals a-b. [2]
05	Determine the Norton equivalent resistance R_N with respect to terminals a-b. [2]

04



$$i_1 = 4\,A$$
$$-2v_x + 2i_2 + v_x = 0$$
$$\text{But} \quad v_x = 1(i_2 - i_1)$$
$$\therefore -2(i_2 - i_1) + 2i_2 + i_2 - i_1 = 0$$
$$i_2 + 8 - 4 = 0$$
$$\therefore i_2 = -4\,A = i_{sc} = I_N$$

05

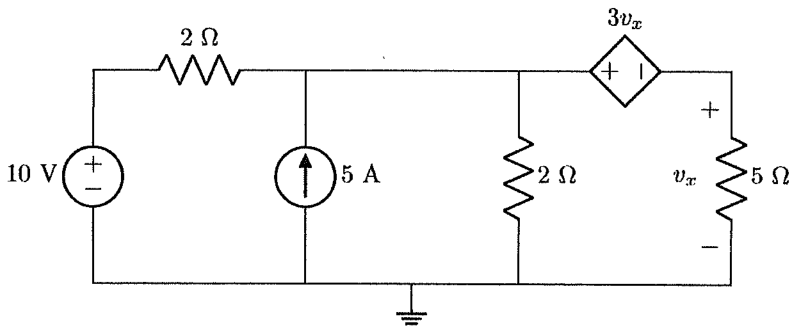


$$\text{KVL}: \quad 2v_x + 2i - v_x - 1 = 0$$
$$v_x = -1i$$
$$\therefore -2i + 2i + i - 1 = 0$$
$$\therefore i = 1\,A$$
$$R_N = \frac{1}{i} = 1\,\Omega$$

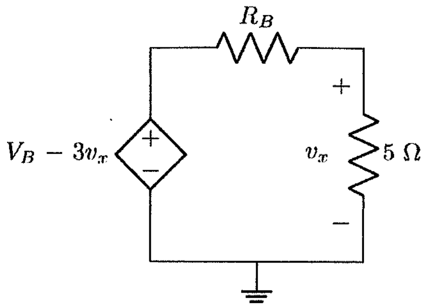
Questions 6 to 7 [3]

Use source transformation to obtain Circuit B so that it is equivalent to Circuit A. Determine R_B and V_B in Circuit B.

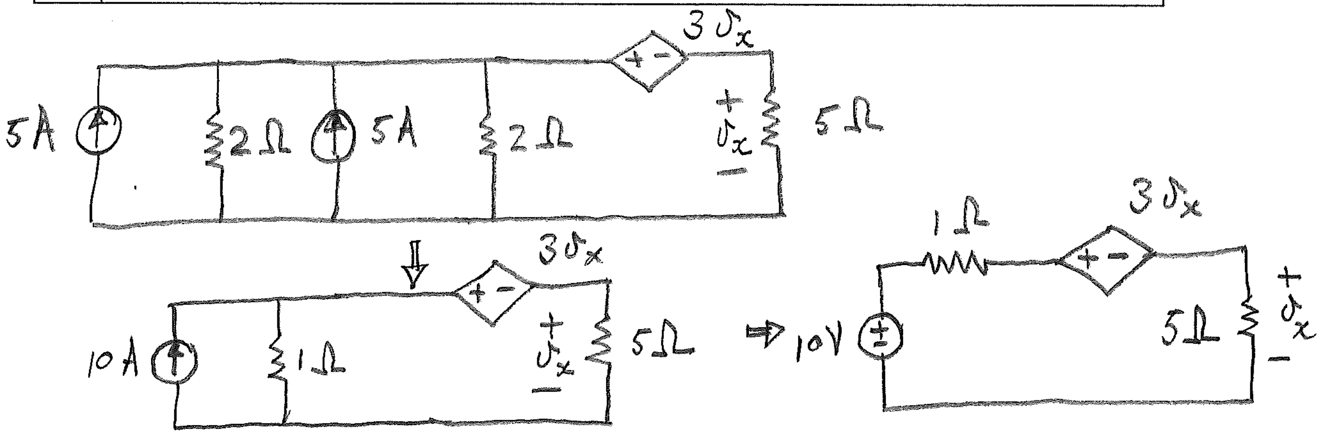
Circuit A:



Circuit B:



06	R_B [1]
07	V_B [2]

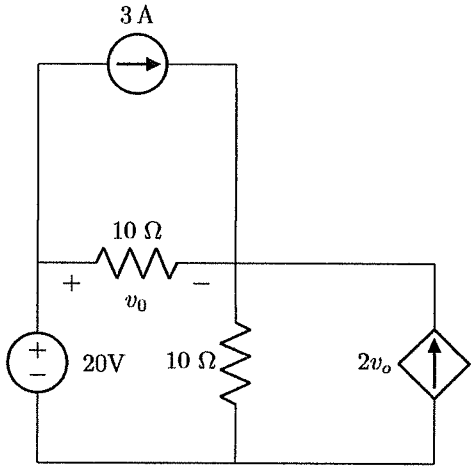


$\therefore\ V_B = 10\ V$

$R_B = 1\ \Omega.$

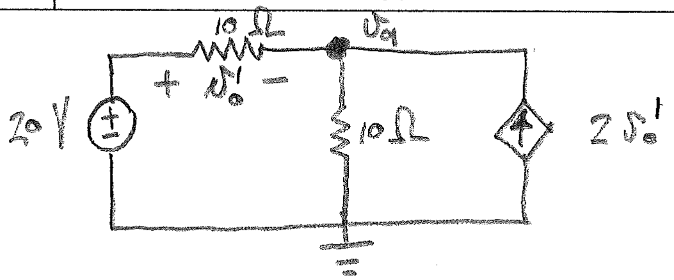
Questions 8 to 9 [4]

Find the individual contributions of each of the 20 V and 3 A sources to the voltage v_o.



08	Contribution of 20 V source [2]
09	Contribution of 3 A source [2]

08

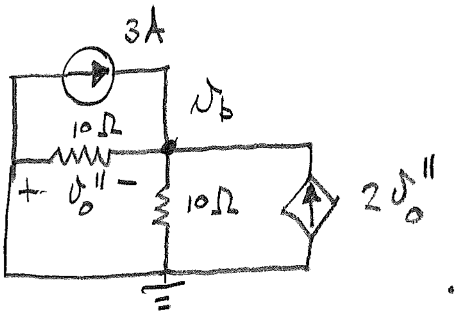


$$\frac{v_a - 20}{10} + \frac{v_a}{10} = 2v_o' \quad \text{But } v_o' = 20 - v_a$$

$$\therefore v_a - 20 + v_a = 400 - 20v_a \quad \therefore v_a = 19.09 \text{ V}$$

$$\text{and } v_o' = 0.91 \text{ V}$$

09



$$\frac{v_b}{10} + \frac{v_b}{10} = +3 + 2v_o''$$

$$v_o'' = -v_b$$

$$\therefore \frac{v_b}{5} + 2v_b = +3 \quad \therefore v_b = 1.364 \text{ V}$$

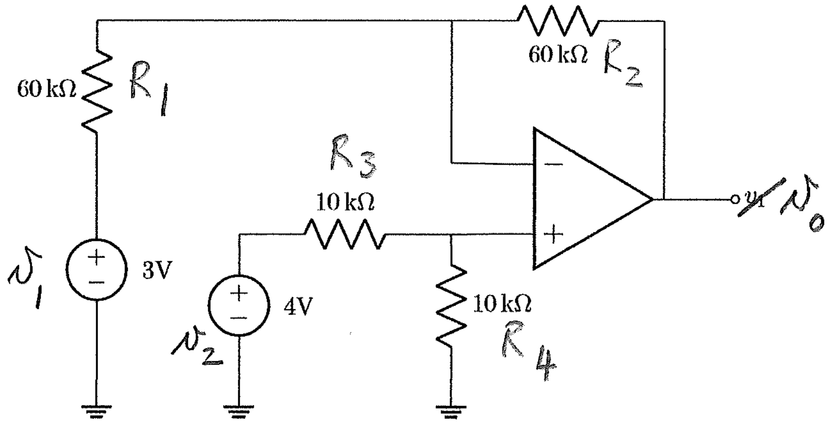
$$\therefore v_o'' = -1.364 \text{ V}$$

Total Section 1: [16]

Section 2

Question 10 [2]

Assume that the op-amp is ideal. Find ~~v_1~~. v_o .



10	v_1 [2]

Since $R_1 = R_2$ and $R_3 = R_4$,

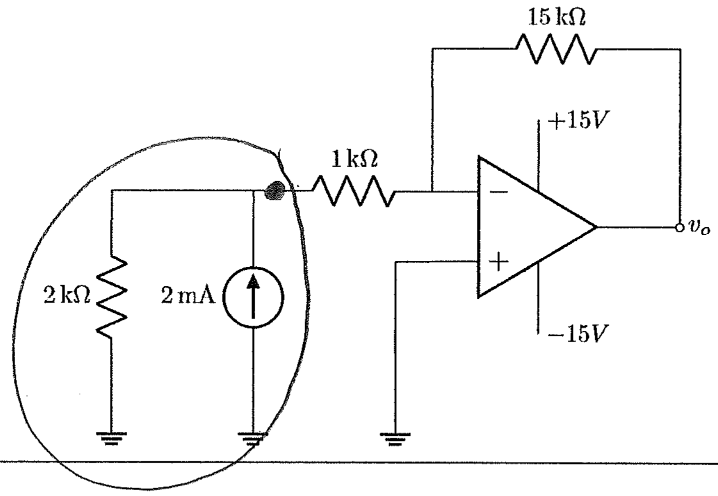
$v_o = v_2 - v_1$ (subtractor)

$= 4 - 3$

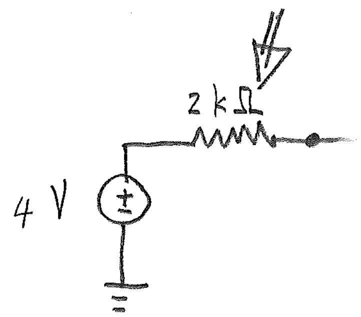
$= 1$ V

Question 11 [2]

Assume that the op-amp is ideal. Find v_o.



11	v_o [2]



Inverting op-amp with $R_1 = 2 + 1 = 3\,k\Omega$.

$$v_o = -\frac{R_f}{R_1} v_i$$

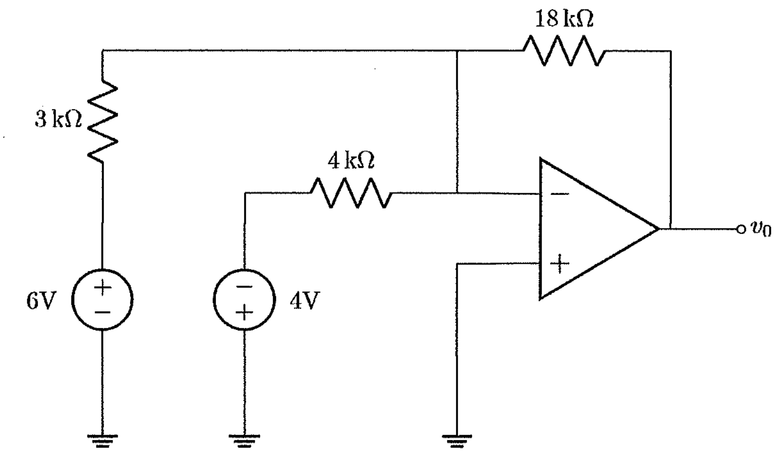
$$= -\frac{15\,000}{3\,000} \cdot 4 = -20\ V$$

But op-amp saturates at $-15\ V$

Hence $v_o = -15\ V$

Question 12 [2]

Assume that the op-amp is ideal. Find v_o.



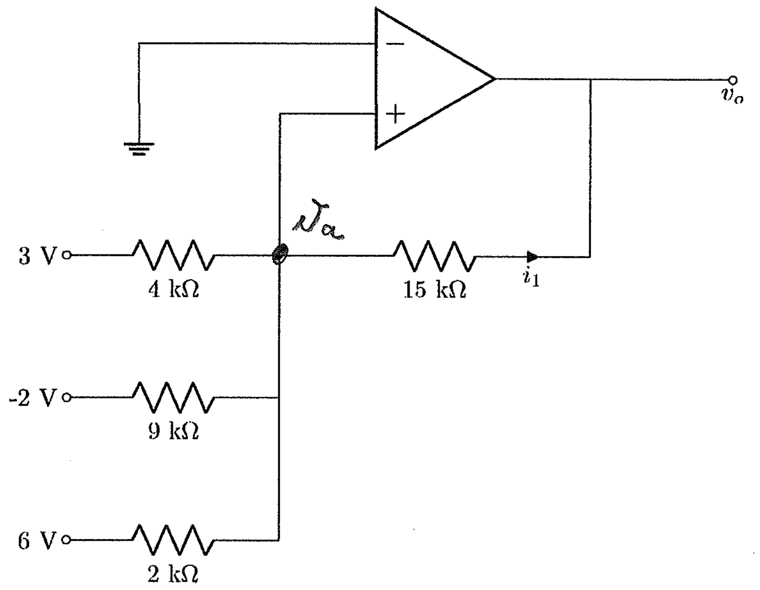
12	v_o [2]

Summing amplifier:

$$v_o = -\left(\frac{18\,000}{3000}6 + \frac{18000}{4\,000}(-4)\right)$$

$$= -18\text{ V}$$

Questions 13 to 14 [4]



13	Calculate v_o. [2]
14	Calculate i_1. [2]

13. $$\frac{v_a - 3}{4000} + \frac{v_a - (-2)}{9000} + \frac{v_a - 6}{2000} + \frac{v_a - v_o}{15000} = 0$$

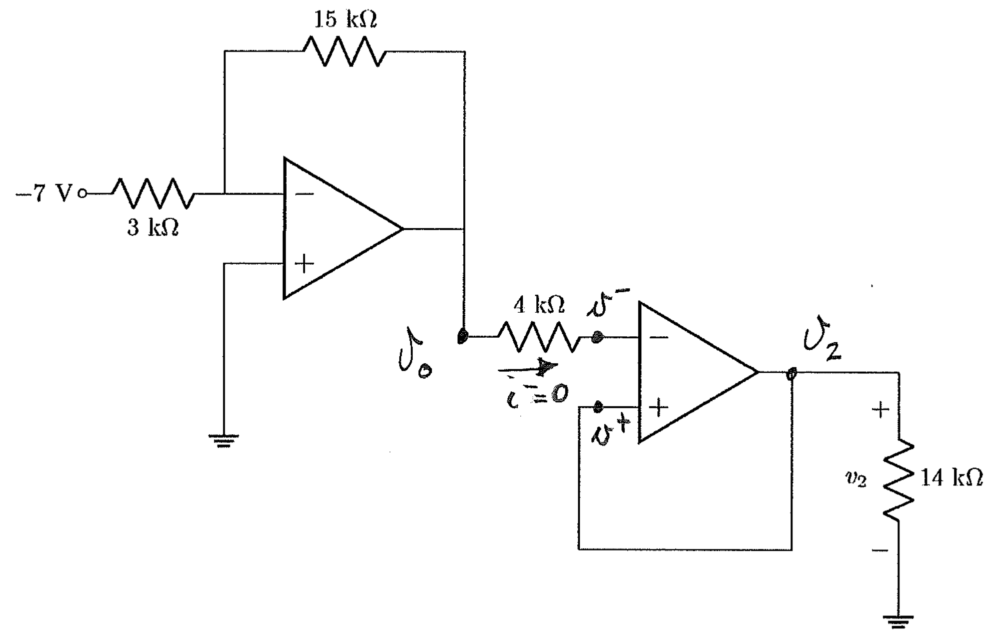
But $v_a = 0$ V

$$\therefore \quad -\frac{3}{4} + \frac{2}{9} - \frac{6}{2} = \frac{v_o}{15}$$

$$\therefore \quad v_o = -52.92 \text{ V}$$

14. $$i_1 = \frac{v_a - v_o}{15000} = 3.528 \text{ mA}$$ (continuity marking)

Question 15 [2]



15	Calculate v_2. [2]

$$v_o = -\frac{R_f}{R_1} v_i$$

$$= -\frac{15\,000}{3000}(-7) = 35 \text{ V} \quad \text{(inverting amplifier)}$$

$v^- = v_o$ since $i^- = 0$

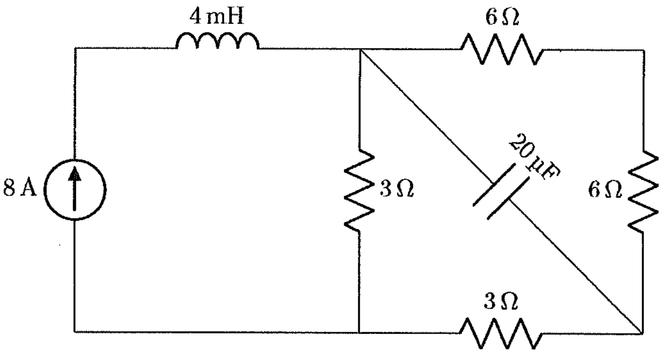
$v^+ = v^-$ (ideal op-amp)

$\therefore v_2 = v^+ = v^- = v_o = 35$ V.

Total Section 2: [12]

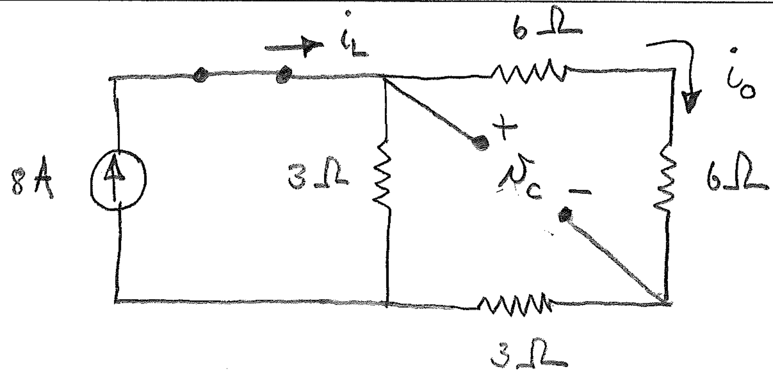
Section 3

Questions 16 to 17 [4]	
Under DC conditions, calculate w_C, the energy stored in the capacitor; and w_L, the energy stored in the inductor.	



16	w_C [2]
17	w_L [2]

DC:

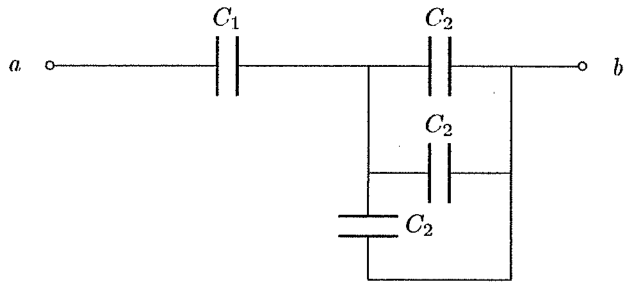


16. $$i_o = 8 \frac{3}{3+(3+6+6)} = 1{,}333 \text{ A}$$

$$v_c = (6+6)\, i_o = 16 \text{ V}$$

$$W_c = \frac{1}{2} C v^2 = \frac{1}{2}(20 \times 10^{-6})\, 16^2 = 2{,}56 \text{ mJ}.$$

17. $$W_L = \frac{1}{2} L i_L^2 = \frac{1}{2}(4 \times 10^{-3})\, 8^2 = 0{,}128 \text{ J}.$$

Question 18 [2]	
Given: $C_1 = 4$ F, $C_2 = 6$ F.	
	
18	Calculate the equivalent capacitance C_{eq} with respect to terminals a - b. [2]

$$C_{\parallel} = C_2 + C_2 + C_2 = 18 \text{ F}$$

$$C_{series} = \frac{C_1 \cdot C_{\parallel}}{C_1 + C_{\parallel}} = \frac{4 \cdot 18}{4 + 18}$$

$$= 3.273 \text{ F}$$

$$= C_{eq}$$

Total Section 3: [6]

Section 4

	Questions 19 to 21 [3]
	The impedance of a circuit element is given by $\mathbf{Z} = 0.5 - j0.4\,\Omega$.
19	Determine the reactance. [1]
20	Determine the admittance. Give your answer in rectangular form. [1]
21	Determine the ~~conductance~~ susceptance. [1]

19. $X = -0.4\ \Omega$

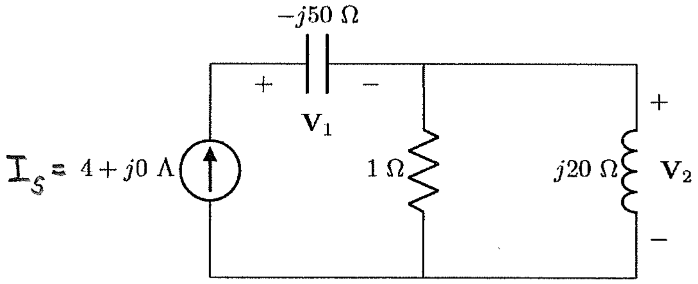
20. $Y = \frac{1}{Z} = 1.220 + j0.9756\ S$

21. $G = \text{Im}\{Y\} = 0.9756\ S$

(continuity marking)

Questions 22 to 23 [4]

Consider the circuit below.



22	Choose the correct option: $\mathbf{I}_s$ and $\mathbf{V}_1$ are in phase (write 0); $\mathbf{I}_s$ leads $\mathbf{V}_1$ (write 1); $\mathbf{I}_s$ lags $\mathbf{V}_1$ (write 2). [2]
23	Choose the correct option: $\mathbf{I}_s$ and $\mathbf{V}_2$ are in phase (write 0); $\mathbf{I}_s$ leads $\mathbf{V}_2$ (write 1); $\mathbf{I}_s$ lags $\mathbf{V}_2$ (write 2). [2]

22. $V_1 = 4 \cdot (-j50) = 200 \angle -90^\circ$ V

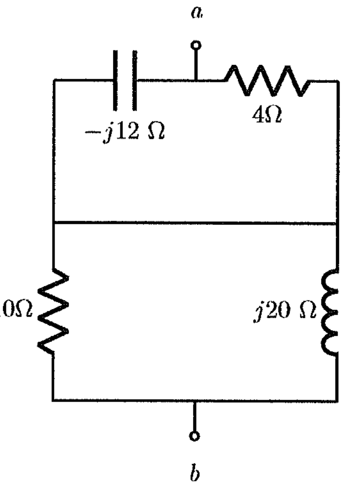
$I_s = 4 \angle 0^\circ$ A

$\therefore$ I_s leads V_1 (1)

23. $V_2 = 4 \left(\frac{1}{1 + j20} \right) \cdot j20$

$= 3.995 \angle 2.862^\circ$ V

$\therefore$ I_s lags V_2 (2).

Question 24 [2]
Determine Z_{ab}. Give the answer in **rectangular form**.


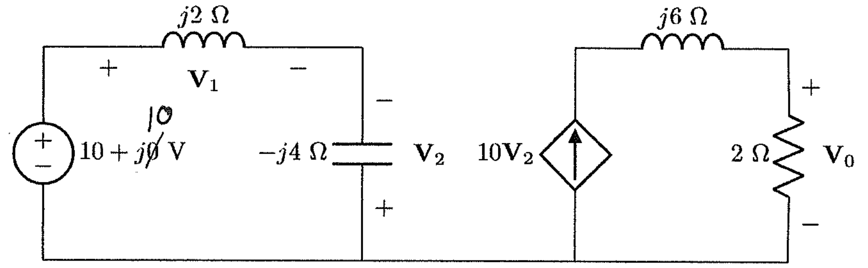
24	Z_{ab} [2]

$$Z_{ab} = -j12 // 4 + 10 // j20$$

$$= \frac{-j12 \cdot 4}{(-j12 + 4)} + \frac{10 \cdot j20}{10 + j20}$$

$$= 11.60 + j2.800 \ \Omega$$

Questions 25 to 26 [3]	
Find the voltages $\mathbf{V}_1$ and $\mathbf{V}_0$. Give your answers in rectangular form.	
25	$\mathbf{V}_1$. [1]
26	$\mathbf{V}_0$. [2]



25 $$V_1 = \frac{j2}{j2 - j4}(10 + j10)$$
$$= -10.0 - j10.0 \text{ V}$$

26 $$V_0 = 2 \cdot 10\, V_2$$
$$V_2 = -(10 + j10)\frac{-j4}{j2 - j4} = -20.0 - j20.0 \text{ V}$$
$$\therefore V_0 = -400 - j400 \text{ V}$$

This page is intentionally left open for rough work.

Total Section 4: [12]

Grand Total : [46]